

LOW ACRYLONITRILE NBR PATENT LANDSCAPE SYNTHESIS

1. ABSTRACT

This report synthesizes a small patent set relevant to low acrylonitrile nitrile-butadiene rubber (NBR) and adjacent rubber-polymer technologies. The supplied evidence does not include extracted monomer ratios, polymerization recipes, or operating conditions for any of the five patents, so the analysis is necessarily limited to title-level and bibliographic signals rather than detailed formulation reconstruction. Within the manifest, the patents span 2009 to 2019 and include one patent explicitly centered on a styrene acrylonitrile isoprene triblock copolymer, two patents focused on polymer preparation or polymerization with living characteristics, one patent on a one-pot synthesis of nanoparticles and liquid polymer for rubber applications, and one broader patent on polymer, process and composition. Collectively, these titles indicate that the relevant technical space is not restricted to conventional emulsion NBR alone, but extends to block copolymer architectures, controlled/living polymerization concepts, and integrated process routes for rubber-compatible polymers and compositions. For low acrylonitrile NBR specifically, the evidence suggests a broader innovation trend toward tailoring elastomer performance through compositional control and process integration, rather than through a single disclosed low-ACN recipe. However, because no structured parameters were extracted, no patent-specific claims can be made regarding acrylonitrile content ranges, monomer feed ratios, initiator systems, emulsifiers, chain-transfer agents, temperatures, pressures, conversion levels, reaction times, or coagulation methods. The most defensible conclusion from the supplied data is that the patent landscape around low acrylonitrile NBR is technologically adjacent to controlled polymerization and composition/process patents, implying that synthesis strategies likely emphasize architecture control, process simplification, and property tuning. Any practical synthesis recommendation must therefore remain general: low acrylonitrile NBR development should prioritize careful monomer-ratio control, robust polymerization process management, and downstream composition optimization, while recognizing that the present evidence set does not support a more specific operating window.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7592409B2

- Patent Title: Styrene acrylonitrile isoprene triblock copolymer
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US9809780B2
- Patent Title: Process for preparing polymers and compositions thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 3

- Patent Number: US7829624B2
- Patent Title: One-pot synthesis of nanoparticles and liquid polymer for rubber applications
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: US10336853B2
- Patent Title: Polymer, process and composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

Patent 5

- Patent Number: US7666962B2
- Patent Title: Polymerization with living characteristics
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

Relevant Experimental Evidence - Example Extracted Examples 1:

Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- All five patents are primary, but the supplied extraction contains no structured monomer, process, or formulation parameters, so direct comparison of acrylonitrile content, ratios,

initiators, emulsifiers, chain-transfer agents, temperatures, pressures, conversions, reaction times, or coagulation methods is not possible from the evidence provided.

- The patent titles indicate a mix of architectures and process themes: US7592409B2 is explicitly a styrene acrylonitrile isoprene triblock copolymer, while US9809780B2, US10336853B2, and US7666962B2 emphasize polymer/process or living-characteristics concepts, and US7829624B2 focuses on one-pot synthesis for rubber applications.
- Historically, the 2009–2010 patents suggest interest in block copolymer and integrated synthesis approaches; the 2017–2019 patents appear to continue the broader trend toward process and composition control, but the evidence does not allow a determination of whether these later patents are more specific to low-acrylonitrile NBR chemistry.
- Recurring approach across the set: polymer architecture and process design are central themes. Key differences cannot be quantified because no extracted experimental parameters were provided.

4. CONCLUSION

Based on the supplied patent evidence, the low acrylonitrile NBR landscape is best characterized as an adjacent set of polymer-technology patents emphasizing architecture control, process integration, and composition tuning rather than a fully disclosed low-ACN formulation space. No patent-specific synthesis window can be reconstructed from the extraction data, so claims about exact acrylonitrile levels, polymerization conditions, or workup methods would be unsupported. The most defensible recommendation is to treat low acrylonitrile NBR development as a controlled polymerization problem: maintain tight monomer-feed control, select a process route compatible with the desired polymer architecture, and optimize downstream composition and processing for rubber performance. The evidence supports a general trend toward more sophisticated polymer/process design, but not a specific operating recipe.

5. REFERENCES

- US7592409B2 | Styrene acrylonitrile isoprene triblock copolymer | Not disclosed | US | 2009 | <https://patents.google.com/patent/US7592409B2/en>
- US9809780B2 | Process for preparing polymers and compositions thereof | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9809780B2/en>
- US7829624B2 | One-pot synthesis of nanoparticles and liquid polymer for rubber applications | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7829624B2/en>
- US10336853B2 | Polymer, process and composition | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10336853B2/en>
- US7666962B2 | Polymerization with living characteristics | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7666962B2/en>